11 Chemistry
Unit 2 Topic Test 3: Gases I (Greenhouse Gases and pV = nRT)

Details	
Student Name:	Teacher Name:
Date Set: ____/____/26	Due Date: ____/____/26

Curriculum Assessed

On completion of Unit 2 the student should be able to measure and analyse chemicals in water and in the atmosphere. This topic test assesses carbon dioxide, methane and water vapour as greenhouse gases, gas pressure in terms of particle collisions, standard laboratory conditions (SLC), and calculations with the universal gas equation, pV = nRT.

Task Overview

Section A – Multiple Choice: 10 questions, 1 mark each (10 marks). Answer on the separate multiple-choice answer sheet.
Section B – Short Answer: 4 questions (30 marks). Answer in this booklet.

Conditions

- Test conditions (revision topic test)
- VCE Chemistry Data Book provided
- Permitted materials: pens, pencils, highlighters, erasers, sharpeners, rulers and one scientific calculator (no CAS).
- Not permitted: notes of any kind, blank paper, white-out, mobile phones, smart watches or any other electronic device.
- 46 mins writing time (about 1.15 minutes per mark)

Marking

- 40 marks
- Marked according to teacher-made marking guide.
- Show all working in calculation questions; give final answers with units and an appropriate number of significant figures. Equations must be balanced and include states.

A+	A	B+	B	C+	C	D+	D	E+	E	NS
90-100%	80-89%	75-79%	70-74%	65-69%	60-64%	55-59%	50-54%	45-49%	40-44%	<40%

Total Marks:		Percentage:		Letter Grade:	

SECTION A – Multiple-choice questions (10 marks)

Answer **all** questions on the separate multiple-choice answer sheet by filling in the letter of the best response. Each question is worth 1 mark. No marks are deducted for incorrect answers. Do **not** write your Section A answers in this booklet.

Question 1

At standard laboratory conditions (SLC), the pressure expressed in atmospheres (atm) is closest to

A. 1.00 atm

B. 1.01 atm

C. 1.01×10^4 atm

D. 0.987 atm

Question 2

The temperature inside a kitchen freezer is −18 °C.

This temperature, in kelvin, is

A. 291 K

B. 255 K

C. −291 K

D. 18 K

Question 3

Water vapour, $H_2O(g)$, is a greenhouse gas because water molecules absorb

A. ultraviolet radiation.

B. visible light.

C. infrared radiation.

D. radio waves.

Question 4

An inflated beach ball keeps a round shape because the air inside pushes equally on every part of its inner surface.

The pressure acts equally in all directions because the air particles

A. move randomly in all directions and collide with every part of the inner surface.

B. are evenly spaced and stay still, pressing outwards on the surface.

C. are pulled down by gravity, so they push hardest on the bottom of the ball.

D. repel each other, and this repulsion pushes outwards on the surface.

Question 5

A 400 mL aerosol can contains propellant gas at a pressure of 450 kPa and a temperature of 22 °C.

The amount of gas in the can is closest to

A. 0.985 mol

B. 73.4 mol

C. 7.34×10^{-5} mol

D. 0.0734 mol

Use the following information to answer Questions 6 and 7.

Air bubbles trapped in Antarctic ice show that, for 800 000 years before 1800, the concentration of carbon dioxide, CO_2, in the atmosphere stayed between about 180 and 280 parts per million (ppm). Since 1800 it has risen to about 424 ppm.

Question 6

The main cause of the rise in carbon dioxide concentration since 1800 is

A. more frequent volcanic eruptions.

B. burning fossil fuels.

C. more photosynthesis by plants and algae.

D. more water evaporating from the oceans.

Question 7

Which one of the following is the most convincing evidence from the information that the recent rise is caused by human activity?

A. In about 200 years the level rose well outside its range for the previous 800 000 years.

B. Carbon dioxide is a greenhouse gas.

C. 424 ppm is still less than 0.1% of the atmosphere.

D. Carbon dioxide levels have changed naturally in the past.

Question 8

A neon sign tube contains 3.20×10^{-4} mol of neon, Ne, at a pressure of 1.75×10^{3} Pa and a temperature of 20 °C.

The volume of the tube is closest to

A. 0.445 mL

B. 30.4 mL

C. 445 mL

D. 4.45×10^{5} mL

Question 9

Two identical sealed flasks are at the same temperature. Flask X contains 2.0 g of hydrogen gas, H_2, and flask Y contains 2.0 g of helium gas, He.

Which one of the following is correct?

A. The pressures in the two flasks are equal, because the masses of gas are equal.

B. The pressure in flask X is twice the pressure in flask Y.

C. The pressure in flask Y is twice the pressure in flask X, because helium atoms are heavier.

D. The pressure in flask X is four times the pressure in flask Y.

Question 10

The mass of methane, CH_4, in a 750 mL container at a pressure of 1.20 atm and a temperature of 40.0 °C is closest to

A. 5.54×10^{-3} g

B. 4.39 g

C. 561 g

D. 0.561 g

END OF SECTION A

SECTION B – Short-answer questions (30 marks)

Answer **all** questions in the spaces provided. Show all working in calculation questions. Give final answers with units and appropriate significant figures. Equations must be balanced and include states.

Question 1 (9 marks)

Coal seams contain trapped methane, CH_4. At an underground coal mine in Queensland's Bowen Basin, methane-rich gas is drained from the coal seam before mining and vented to the atmosphere.

The mine releases about 1.20×10^6 L of methane per day, measured at 38.0 °C and 97.5 kPa. The mine plans to capture 80% of this methane and burn it to generate electricity.

a) State one natural source of methane.

1 mark

b) Name one human activity, other than mining, that releases methane into the atmosphere.

1 mark

c) Calculate the mass of methane, in kg, released by the mine in one day.

3 marks

d) Calculate the mass of methane, in tonnes, that capturing 80% of the methane would keep out of the atmosphere in one year (365 days). (1 tonne = 1000 kg)

2 marks

e) Methane from natural sources and methane drained from the coal seam are the same substance, and both formed by natural processes. Explain why the methane from the mine counts as adding to the enhanced greenhouse effect, but methane from natural sources does not.

2 marks

Question 2 (9 marks)

Aisha is studying how her lungs work. At rest, each breath she exhales is 500 mL of air at body temperature, 37.0 °C, and a pressure of 101 kPa. Exhaled air contains 4.0% carbon dioxide, CO_2, by amount (mol).

a) Convert 37.0 °C to kelvin.

1 mark

b) Convert 101 kPa to pascals.

1 mark

c) Calculate the total amount, in mol, of gas in one exhaled breath.

2 marks

d) Aisha breathes 12 times per minute. Calculate the mass of carbon dioxide that she exhales in one day (24 hours).

3 marks

e) When Aisha breathes in, her chest muscles increase the volume of her lungs. Explain, in terms of gas particles, why air flows into her lungs.

2 marks

Question 3 (6 marks)

On a winter morning in Ballarat, a heater warms the air in a classroom from 5.0 °C to 22.0 °C. The classroom has a volume of 2.40×10^5 L, and the air pressure inside stays at 101 kPa as the air is heated. The molar mass of air can be taken as 28.8 g mol^{-1}.

a) State the SI unit of pressure.

1 mark

b) Explain, in terms of the gas particles, why some air must leave the classroom as the air inside is heated.

2 marks

c) Calculate the mass of air, in kg, that leaves the classroom as it is heated.

3 marks

Question 4 (6 marks)

Liam drives a total of 40 km each day, to work and home again, in Melbourne. His car emits 168 g of carbon dioxide, CO_2, for each kilometre travelled.

a) Calculate the mass, in kg, of carbon dioxide emitted by Liam's car each day.

1 mark

b) Calculate the volume of this carbon dioxide at 16.0 °C and 100.5 kPa, in L.

3 marks

c) The same amount of carbon dioxide leaves the exhaust pipe at about 200 °C, at the same pressure. Explain whether the volume of this carbon dioxide is larger or smaller at 200 °C than your answer to part b).

2 marks

END OF SECTION B

Additional space

Clearly number any response written here.

END OF TEST

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SECTION A – MULTIPLE-CHOICE ANSWER SHEET

Student name: _____ Teacher: _____

Shade or circle ONE letter for each question. Each question is worth 1 mark. If you change an answer, cross out the first answer clearly.

Q	A	B	C	D
1	(A)	(B)	(C)	(D)
2	(A)	(B)	(C)	(D)
3	(A)	(B)	(C)	(D)
4	(A)	(B)	(C)	(D)
5	(A)	(B)	(C)	(D)
6	(A)	(B)	(C)	(D)
7	(A)	(B)	(C)	(D)
8	(A)	(B)	(C)	(D)
9	(A)	(B)	(C)	(D)
10	(A)	(B)	(C)	(D)

Section A score: _____ / 10